



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

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TITLE OF THE PROJECT	FoodRescue AI: Autonomous Surplus-Food Detection & Redistribution Agent	
S. No.	University ID	Name
1	2420030749	P. Poojith Varma Sai
2	2420090072	G. Prashanth Kumar
3	2420030020	V. Harshith Chandra Varma
Name of the Guide		Venkateswari

Abstract:

Food wastage is a significant societal problem, particularly in places such as college canteens, restaurants, hotels, hostels, cafeterias, and large events where prepared food may remain unused after consumption periods. A major challenge is that surplus edible food is not always redistributed quickly because identifying the available food, determining its urgency, finding suitable recipients, and coordinating pickup often require manual effort. This coordination gap can result in usable food becoming waste while organizations and communities may simultaneously require food resources.

To address this problem, FoodRescue AI is proposed as an AI-based autonomous surplus-food detection and redistribution agent. The system enables food providers such as canteens, restaurants, hotels, and event organizers to submit information about available surplus food through a web application. The AI Agent analyzes the submitted information and determines important characteristics such as food category, quantity, urgency, and redistribution suitability. Based on these factors, the agent intelligently identifies suitable nearby recipient organizations and determines the priority of redistribution.

The proposed system goes beyond conventional food donation management by introducing an autonomous decision-making and coordination layer. The AI Agent can analyze surplus-food information, recommend an appropriate recipient, prioritize the redistribution request, initiate notifications, coordinate pickup requirements, and monitor the redistribution status. This enables the system to follow an observe–analyse–decide–act–monitor workflow rather than simply storing donation records.

The project introduces three major novelties: AI-based surplus-food intelligence, which analyzes food information and determines urgency and suitability; intelligent recipient matching, which matches available food with suitable recipients based on factors such as food type, quantity, location, and requirement; and autonomous redistribution coordination, where the AI Agent assists in prioritizing requests, initiating notifications, coordinating pickup, and tracking the redistribution process.

The system will be implemented using a Service-Oriented Architecture (SOA) with modular microservices. Core services will include User Management, Surplus-Food Management, AI Agent Processing, Recipient Matching, and Notification/Redistribution Coordination. An API Gateway and service discovery mechanism will provide controlled communication between services, while REST APIs will support service interaction. Authentication and authorization will be implemented to provide secure access to the application. The AI Agent will be integrated as a dedicated service within the overall architecture.

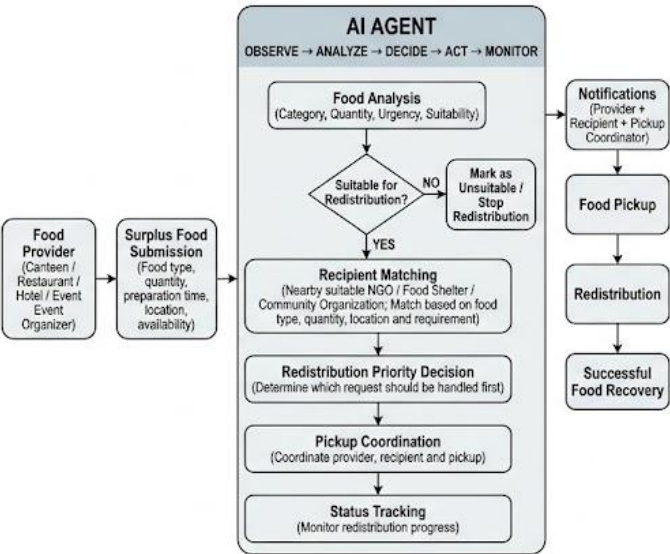
The proposed solution aims to reduce avoidable food wastage, improve the speed of surplus-food redistribution, and provide a scalable digital platform connecting food providers with suitable recipient organizations. The project contributes primarily to SDG 2 – Zero Hunger by supporting the redistribution of usable food to communities and organizations that can utilize it, and SDG 12 – Responsible Consumption and Production by promoting the efficient use of food resources and reducing unnecessary waste.

The expected outcome is a functional web-based application capable of intelligently processing surplus-food information, recommending suitable recipients, prioritizing redistribution, and coordinating the overall workflow through an autonomous AI Agent and SOA-based microservices architecture.

Project Architecture/Project Overview:

SYSTEM WORKFLOW DIAGRAM

'FoodRescue AI: Autonomous Surplus-Food Detection & Redistribution Agent'



Reduced Food Waste + Faster Redistribution + Social Impact

FoodRescue AI – SOA Architecture

